

DR. PANDA AI HEALTHCARE DIAGNOSTIC CHATBOT

System Architecture, Dataset Pipeline, and Latency Evaluation

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Domain: Natural Language Processing

Deployment: FastAPI & Streamlit

PERFORMANCE METRICS

76.8%

Semantic Accuracy

0.89 ms

Query Latency

96.0%

Top Match Confidence

FastAPI

API Framework

PROJECT DEEP DIVE

Dr. Panda is an AI powered medical diagnostic assistant designed to analyze patient symptom descriptions and retrieve screening condition matches with clinical recommendations. The system uses a multi model similarity engine combining TF IDF n gram vectorization and BERT embeddings.

- Primary Dataset: Kaggle Medical Question Answering Dataset
- Local Data Storage: Included in data/medical_qa_dataset.csv
- Response Time: Sub 200 ms production goal achieved with 0.89 ms local engine latency
- API Documentation: Interactive Swagger UI available at /docs

SYMPTOM QUERY MATCH SAMPLE

Query Input	Top Condition Match	Confidence Score	Clinical Recommendation
sharp headache nausea light sensitivity vision blurring	Migraine Headache	96.0%	Rest in a dark quiet room. Hydrate and consider pain relief.
persistent cough shortness of breath fever chest discomfort	Acute Bronchitis	88.5%	Consult a physician for a chest evaluation. Rest and stay hydrated.